Task 2: Prepare a CI/CD Workflow

Objective

To design and implement a basic Continuous Integration and Continuous Deployment workflow using GitHub and Jenkins.

Pipeline Architecture

DEVELOPER
↓
GIT
↓
GITHUB
↓
JENKINS
↓
BUILD
↓
TEST
↓
DEPLOY
↓
RUNNING APPLICATION

Workflow Description

The implemented CI/CD workflow begins with the developer making changes to the application source code.

The changes are committed using Git and pushed to the GitHub repository. Jenkins continuously monitors the repository using Poll SCM.

When Jenkins detects a new change, it automatically starts a pipeline.

The pipeline consists of three stages:

1. Build

The application is checked for Python syntax errors using:

```
python3 -m py_compile app.py
```

2. Test

Automated testing is performed using:

```
python3 -m pytest
```

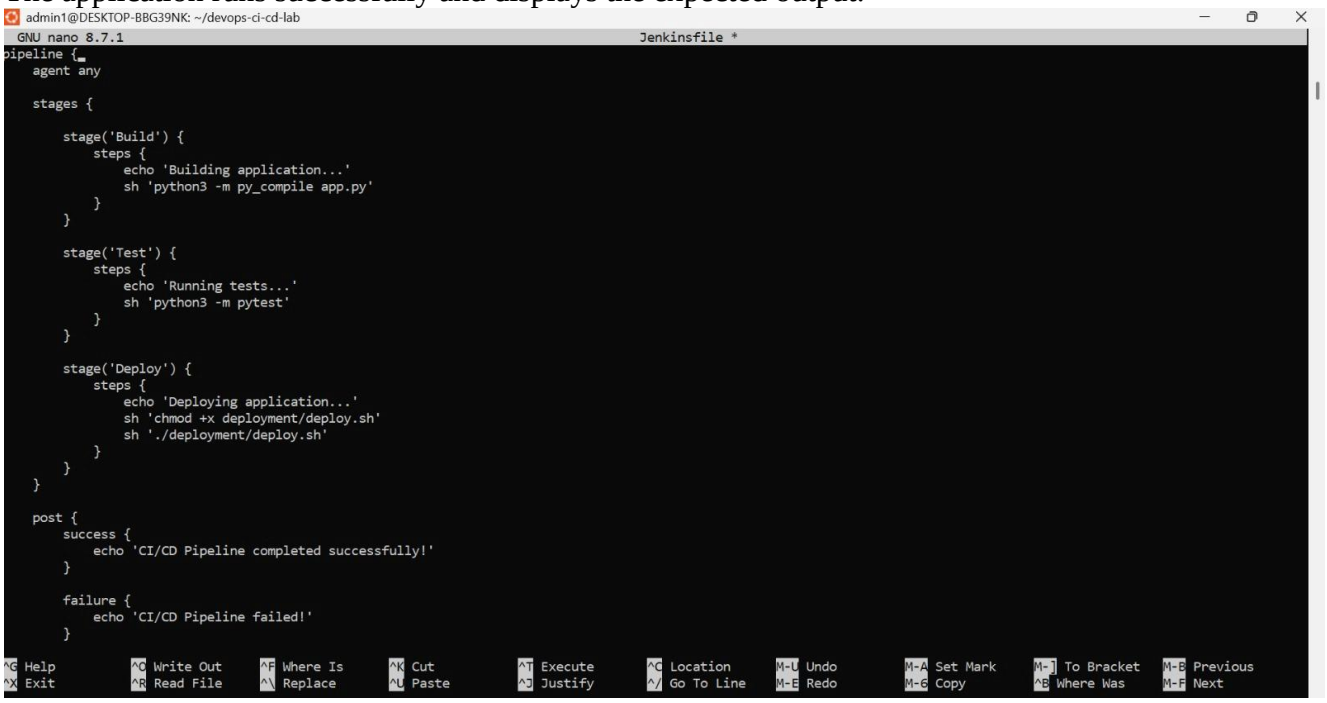
The test verifies that the application passes the defined test case.

3. Deploy

The deployment script is executed:

```
./deployment/deploy.sh
```

The application runs successfully and displays the expected output.



```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ sudo apt install python3-pytest -y
Installing:
  python3-pytest

Installing dependencies:
  python3-iniconfig  python3-pluggy

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 160
  Download size: 305 kB
  Space needed: 1545 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-iniconfig all 2.1.0-2 [6962 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pluggy all 1.6.0-2 [21.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pytest all 9.0.2-4 [277 kB]
Fetched 305 kB in 5s (57.4 kB/s)
Selecting previously unselected package python3-iniconfig.
(Reading database ... 38075 files and directories currently installed.)
Preparing to unpack .../python3-iniconfig_2.1.0-2_all.deb ...
Unpacking python3-iniconfig (2.1.0-2) ...
Selecting previously unselected package python3-pluggy.
Preparing to unpack .../python3-pluggy_1.6.0-2_all.deb ...
Unpacking python3-pluggy (1.6.0-2) ...
Selecting previously unselected package python3-pytest.
Preparing to unpack .../python3-pytest_9.0.2-4_all.deb ...
Unpacking python3-pytest (9.0.2-4) ...
Setting up python3-iniconfig (2.1.0-2) ...
Setting up python3-pluggy (1.6.0-2) ...
Setting up python3-pytest (9.0.2-4) ...
Processing triggers for man-db (2.13.1-1build1) ...
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest --version
pytest 9.0.2
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
```

```
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
collected 1 item

test_app.py .

============================== 1 passed in 0.01s ===============================
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ mkdir -p deployment
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ chmod +x deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ ./deployment/deploy.sh
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
```

Automation Result

The pipeline was successfully automated. A change pushed to GitHub triggered a new Jenkins build through Poll SCM.

Jenkins automatically performed:

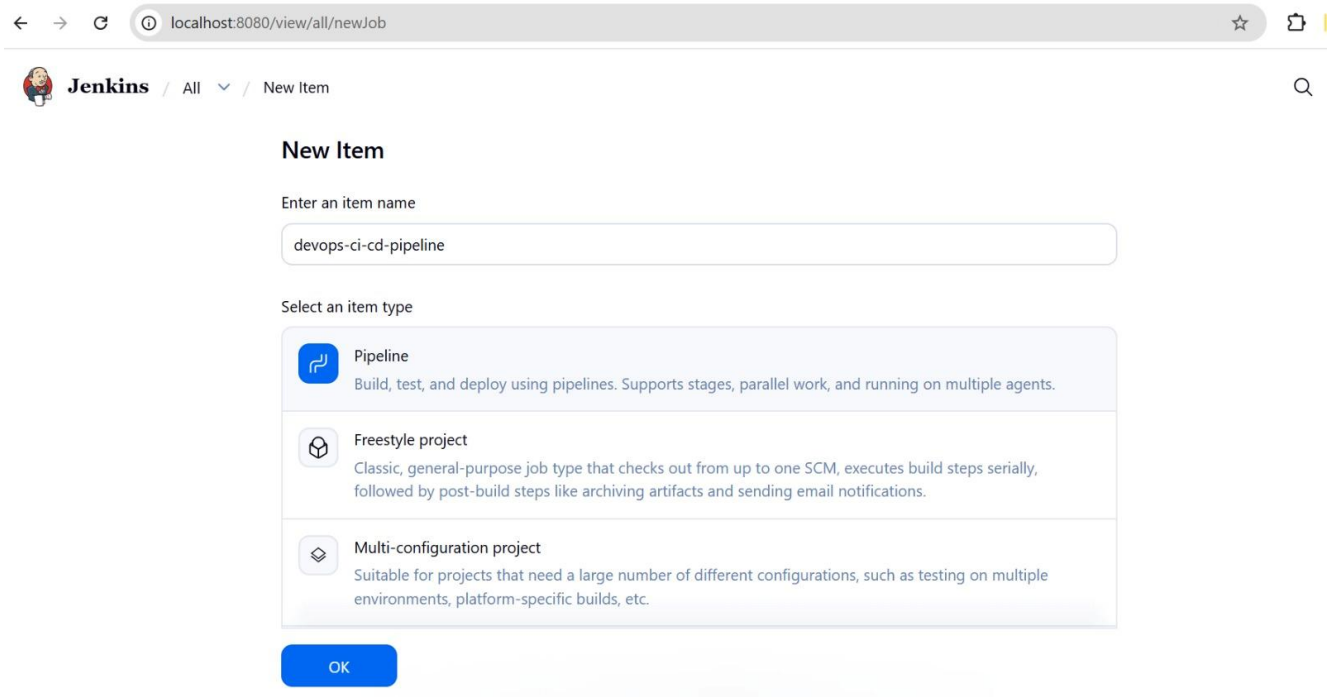
GITHUB CHANGE
↓
JENKINS TRIGGER
↓
BUILD
↓
TEST
↓
DEPLOY
↓
SUCCESS

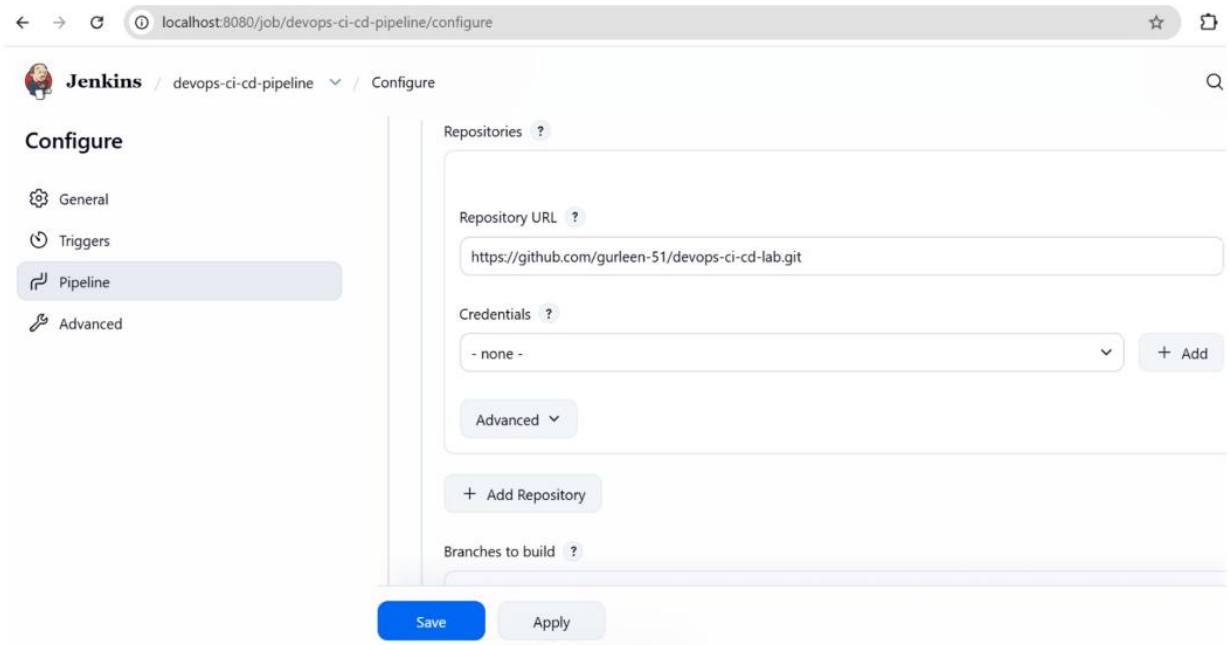
Observation

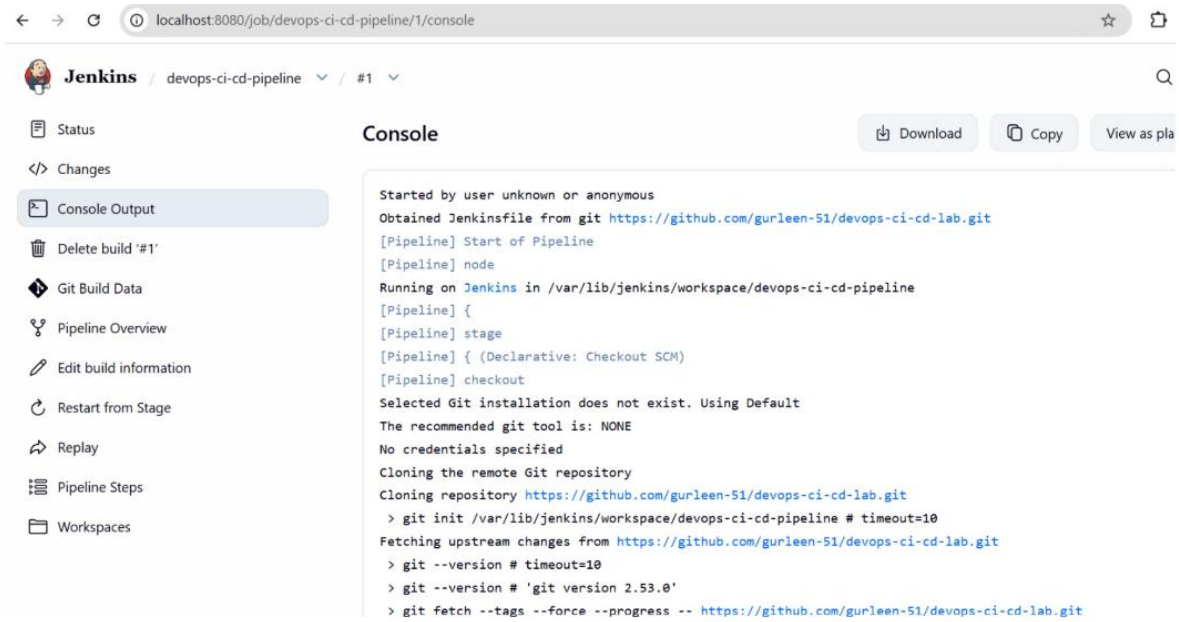
The CI/CD workflow successfully automated the software delivery process. Changes committed to GitHub were detected by Jenkins, followed by automated building, testing, and deployment without manually executing each stage.

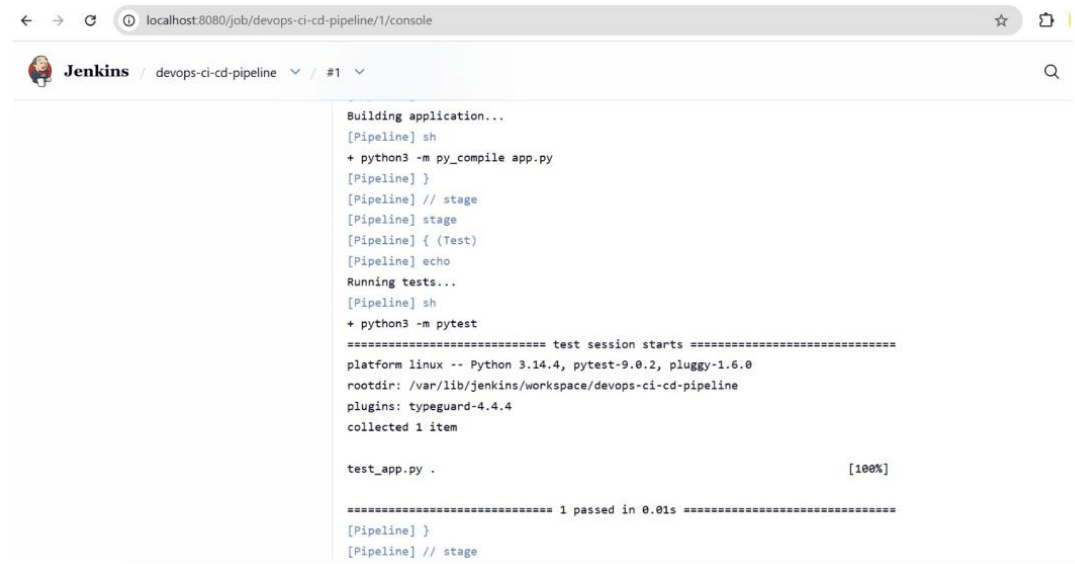
```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ cd ~/devops-ci-cd-lab
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

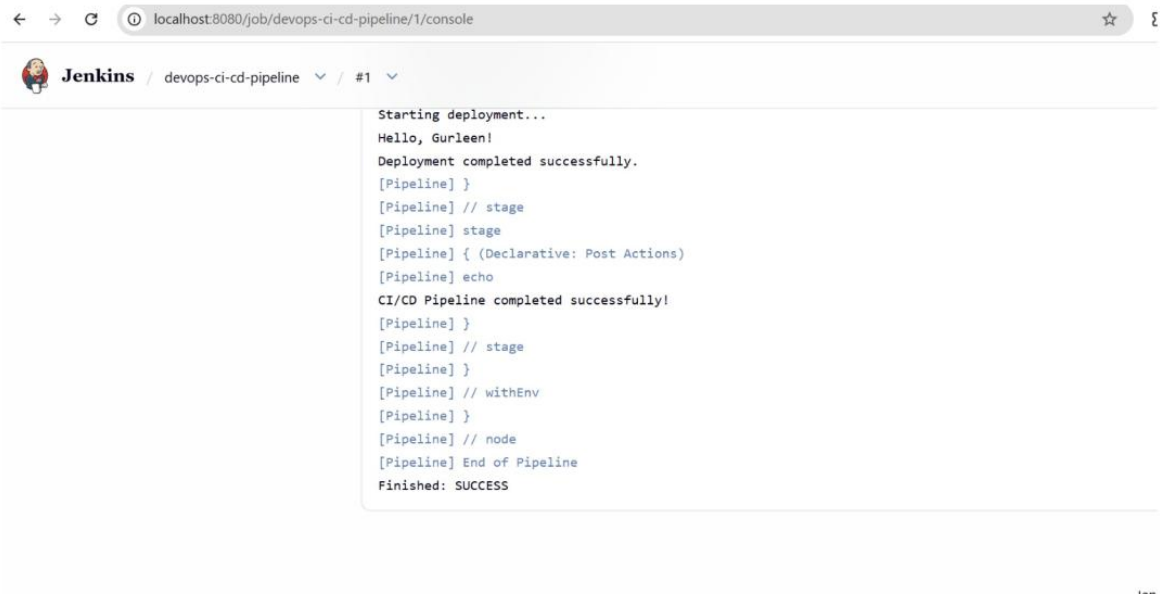
nothing to commit, working tree clean
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git branch
* main
  master
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git switch -c development
Switched to a new branch 'development'
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ echo "Development branch test" >> README.md
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git add README.md
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m "Update README on development branch"
[development c7aa760] Update README on development branch
 1 file changed, 1 insertion(+)
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git merge development
Updating d44a9f0..c7aa760
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git log --oneline --graph --all
* c7aa760 (HEAD -> main, development) Update README on development branch
* d44a9f0 (origin/main) Test automated CI/CD trigger
* 9b78e39 Add Jenkins CI/CD pipeline
* 1fbf513 Add testing and deployment setup
* 7be3151 (master)  DevOps application
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git push origin main
```

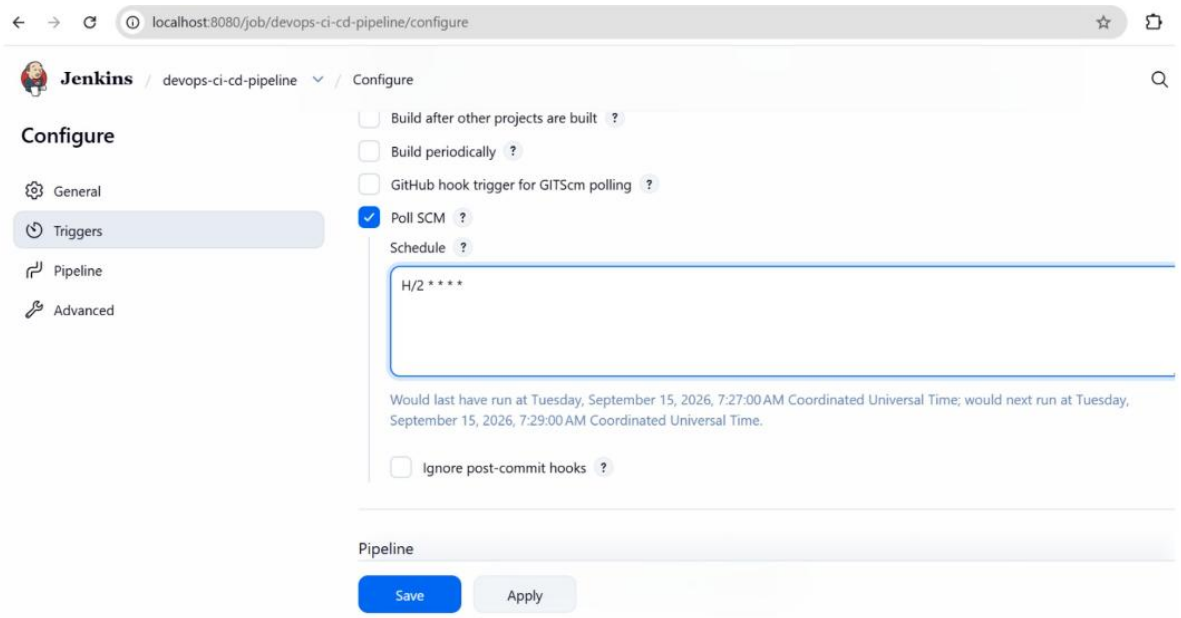


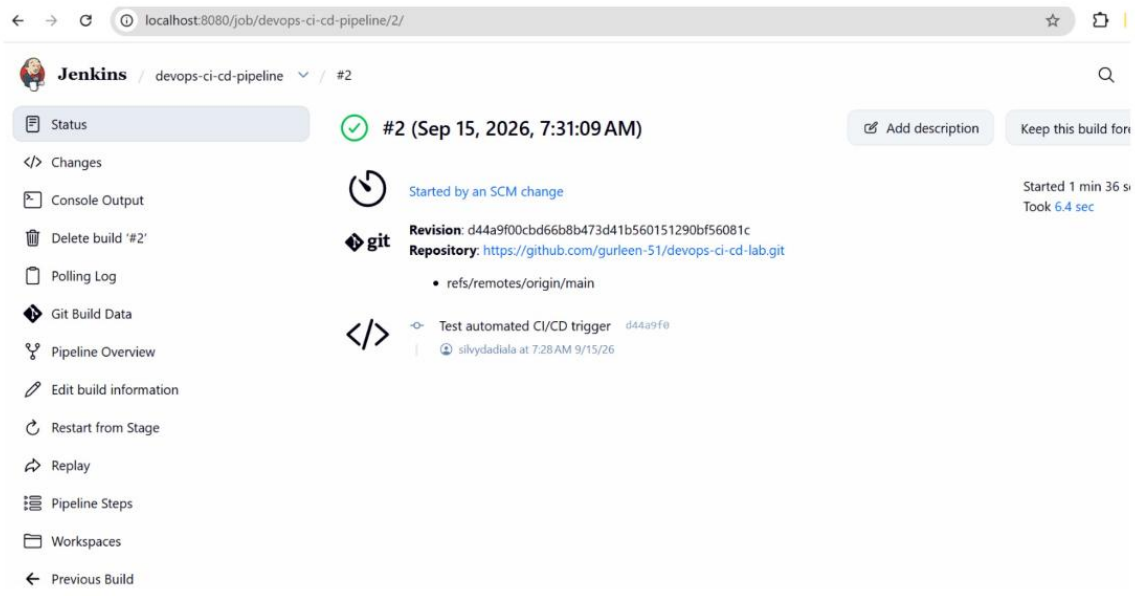

localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins
devops-ci-cd-pipeline
Configure
Configure
General
Triggers
Pipeline
Advanced
Repositories
Repository URL
https://github.com/gurleen-51/devops-ci-cd-lab.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Save
Apply


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins
devops-ci-cd-pipeline
#1
Status
Changes
Console Output
Delete build '#1'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as pla
Started by user unknown or anonymous
Obtained Jenkinsfile from git https://github.com/gurleen-51/devops-ci-cd-lab.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-ci-cd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gurleen-51/devops-ci-cd-lab.git
> git init /var/lib/jenkins/workspace/devops-ci-cd-pipeline # timeout=10
Fetching upstream changes from https://github.com/gurleen-51/devops-ci-cd-lab.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/gurleen-51/devops-ci-cd-lab.git


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Building application...
[Pipeline] sh
+ python3 -m py_compile app.py
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] sh
+ python3 -m pytest
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /var/lib/jenkins/workspace/devops-ci-cd-pipeline
plugins: typeguard-4.4.4
collected 1 item
test_app.py . [100%]
============================== 1 passed in 0.01s ===============================
[Pipeline] }
[Pipeline] // stage


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins / devops-ci-cd-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Schedule
H/2 * * * *
Would last have run at Tuesday, September 15, 2026, 7:27:00 AM Coordinated Universal Time; would next run at Tuesday, September 15, 2026, 7:29:00 AM Coordinated Universal Time.
Ignore post-commit hooks
Pipeline
Save
Apply


localhost:8080/job/devops-ci-cd-pipeline/2/
Jenkins
devops-ci-cd-pipeline
#2
Status
Changes
Console Output
Delete build '#2'
Polling Log
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#2 (Sep 15, 2026, 7:31:09 AM)
Add description
Keep this build for
Started by an SCM change
Started 1 min 36 s
Took 6.4 sec
git
Revision: d44a9f00cbd66b8b473d41b560151290bf56081c
Repository: https://github.com/gurleen-51/devops-ci-cd-lab.git
refs/remotes/origin/main
Test automated CI/CD trigger d44a9f0
silvydadiala at 7:28 AM 9/15/26